

F24-200-R-INS(P)ECT Optimized Flea Beetle Counting using Edge Devices

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Chapter 1

Introduction

This report outlines the methodology employed to research the development and optimization of a deep learning AI model. The aim of this model is to detect flea beetles on pictures of sticky traps, while being deployed on edge devices such as mobile phones. The objectives of this study are to compare previous models, improve or maintain the accuracy of detecting flea beetles, minimize energy consumption on edge devices, and enhance the collaboration with external institutes and to contribute to the field of edge-AI in architecture.

The problem we intend to solve in this study is the lack of presence of optimized, energy-efficient, readily available and most of all accurate methods of automated insect detection for farmers. We identified the research gaps in previous studies pertaining to the solutions of pest detection. No automated solutions currently exist which do not require specialized hardware, or which are commercially available. The goal of our research is to create a one-stop solution to bridge all the gaps we identified.

Our research will focus on identifying the techniques required to develop an AI model which accomplishes the goal of insect detection efficiently. We will then optimize the model for use on edge devices, maintaining the accuracy while reducing the energy consumption and inference time. The AI model, integrated into a mobile application, will serve as a cost-friendly and readily available solution to the problem of pest detection in agriculture.

1.1 Problem Domain

We realized that the agricultural sector is faced with challenges in the form of pest detection. The challenges are primarily identified as accurate and timely detection of pest populations. The traditional method for estimating the population of pests involves counting them manually after catching them on a sticky trap. Due to human error, this process

can be error prone leading to excessive usage of insecticide. The alternative is automated solutions to remove the labor and inaccuracy attached to traditional methods. However, these automated solutions are not accessible for small-scale farmers as they require specialized equipment and setup.

With the use of AI on edge devices, we can revolutionize the field of pest detection in agriculture. Our solutions proposes to remove the problems of availability, complexity and manual labor providing an AI model optimized for use on edge devices such as mobile phones allowing pest detection and insecticide planning can be efficiently taken care of by taking a picture of a sticky trap on your mobile phone.

1.2 Research Problem Statement

Extensive research has been done on small object detection and specifically on insect detection as well. However, these solutions require specialized hardware and can be expensive and difficult to implement for small-scale farmers. The automated solution that our collaborator Dr. Abdul Bais' team developed in [10] is the Smart Insect Trap (SIT). It is equipped with a camera which takes pictures of a sticky trap and uses a deep learning model to determine the count of insects. This result is then uploaded to a cloud server for analysis. However, this solutions is expensive and inaccessible requiring the use of NVIDIA Jetson Nano GPU.

Our research focuses on providing a readily accessible and user-friendly solution to automate the process of pest detection. We will focus on the techniques to develop an AI model capable of accurate pest detection with low energy consumption on edge devices. We will then optimize this model for integration with edge devices so that it decreases the inference time required while providing accurate counts for optimal planning of insecticide use.

1.3 Software Requirements Specification

This research requires the development of a mobile application to showcase the results we will accomplish. The mobile application will cover the following aspects.

1.3.1 Functional Requirements

- The application will allow users to capture images of sticky traps which can be analyzed for the detection of flea beetles.

- The application will have an AI model process these images to accurately identify and count the flea beetles captured on the sticky trap.
- The application will have a user-friendly and intuitive user interface for ease of navigation and use.
- The application will generate reports for the analysis of the population of flea beetles present in the field and provide information about the trend of population increase or decrease
- The application will allow these reports to be stored on the cloud where they can be easily accessed and stored.

1.3.2 Non-functional Requirements

- The application will inference the model to provide the results in a minimal time span.
- The application will be optimized for low battery consumption to prolong battery life.
- The application will have a user-friendly and intuitive user interface for ease of navigation and use.
- The application will be compatible with a range of mobile devices to ensure wide availability
- The application will handle data securely, only allowing access to authorized users.

1.3.3 Features

Following are the features that the mobile application will have:

1. **Image Capture and Upload:** The app will allow users to capture images of insects on sticky traps.
2. **Automatic Insect Counting:** The app will provide this image to the AI model for detection and counting.
3. **Edge-AI:** The app will be integrated with the AI model and the model will be inferred on the mobile device itself.
4. **Easy-to-use User Interface:** The app will have an easy and user-friendly interface.

5. **Offline Functionality:** The app will be able to perform the detection and counting offline as well.
6. **Results Summary and History:** The app will compile the results of the inference and keep a track of the history.
7. **Data Visualization:** The app will visualize the data for the user for better understanding.
8. **Cloud Backup:** The app will backup the data to the cloud for storage .

Chapter 2

Literature Review

In this section, we will perform a critical analysis of existing research in regards to the problem domain. Note that we have reviewed multiple research papers, but only the five most relevant ones are discussed in detail below.

2.1 Related Research

Following are the related research items we will be discussing in detail.

2.1.1 Research Item 1

A novel computer vision system for efficient flea beetle monitoring in canola crop by Ullah et al. [10]

2.1.1.1 Summary of the research item

This research paper presents a novel insect-monitoring system for canola crops that addresses the challenges of manual monitoring for flea beetles (FBs). The proposed solution involves a solar-powered smart insect trap (SIT) equipped with a high-resolution camera and a deep-learning-based detection network called FB-YOLO, which is a modified version of YOLOv8s. By enhancing the network's neck to better aggregate features from different layers, FB-YOLO improves the detection of small FBs, achieving an average precision (mAP@0.5) of 89.97%, which is a 1.215% improvement over the standard YOLOv8s. Additionally, the integration of a confluence non-max suppression algorithm (NMS-C) to reduce duplicate detections further boosts the mAP@0.5 to 90.16%. This system enables hourly monitoring and rapid detection of FB population increases, enhancing crop health management.

2.1.1.2 Critical analysis of the research item

Some of the strengths of the paper include the introduction of a new object detection model for the SITs, designed for flea beetle detection. The model is a modification of the YOLOv8 architecture. The paper works on a comprehensive custom dataset collected specifically for automating the processes of insect counting. The classification of images based on FB density (sparse, medium, and dense) allows for targeted evaluation of model performance. The paper also uses a special class-wise version of the non-max suppression algorithm for better accuracy as opposed to the default one. Overall, the SIT is a neat solution. However, there are some limitations as well. Firstly, the SITs are expensive embedded systems containing hardware such as a high resolution camera and the Nvidia Jetson Nano GPU. Apart from being expensive, these solutions are not easily accessible to farmers. Secondly, there is a dependence on a cloud server for analysis, ruling out the SITs from being true edge-devices. Lastly, the convolutional neural network (CNN) used within the SITs struggles with integration on other platforms. Hence, there is a need for research on developing an optimized beetle counting model for mobile phones.

2.1.1.3 Relationship to the proposed research work

The paper serves as the primary basis for our research, providing valuable insights into automated pest counting methods. We aim to address the gaps highlighted with our own research, offering a more reliable solution to farmers. The current solution presented in the paper is not only too expensive, but less readily available. Additionally, the reliance on server-based processing further limits usability, especially in remote agricultural settings. Our goal is to bring the solution to readily accessible mobile devices, allowing farmers to utilize modern technology in their practices without having to go out of their way to buy specialized equipment. For this purpose, we are collaborating with experts like Dr. Abdul Bais and partnering with Agriculture and Agri-Food Canada, ensuring that we fill the gaps identified in the paper and contribute to the advancement of agricultural practices.

2.1.2 Research Item 2

A vision-based counting and recognition system for flying insects in intelligent agriculture by Zhong et al. [12]

2.1.2.1 Summary of the research item

The paper presents a vision-based counting system designed to detect and classify flying insects. The implementation of the system features the YOLO (You Only Look Once) [6]

model for detection, and SVM (Support Vector Machine) for classification. The methodology involves using deep learning techniques for counting, and SVM for identification between six insect species based on features such as color, texture, and shape. An average counting accuracy of 92.50% and a classification accuracy of 90.18% were observed. The system operates efficiently on a Raspberry PI, allowing for real-time monitoring of pest populations.

2.1.2.2 Critical analysis of the research item

The paper highlights significant advancements in pest detection using deep learning. One of the paper's strengths lies in its ability to handle insufficient training data. In the first phase, all flying insects are regarded as one class and are detected via deep learning. In the second phase, SVM is used for fine classification. Another strength is the use of yellow sticky traps in the wild for image collection, rather than in a controlled lab environment, demonstrating the paper's real-world application. The system runs on a Raspberry Pi enabling edge-computing which is another notable strength. However, there are some limitations as well. Firstly, the system's reliance on a limited dataset raises questions about generalizability. Secondly, while the system runs on edge, there is still a reliance on a cloud server for analysis.

2.1.2.3 Relationship to the proposed research work

The paper presents a vision-based system for detecting insects, which shares a common goal with our proposed work of enhancing pest management through technological innovation. Both our work and the paper rely on yellow sticky traps for pest detection. Our research specifically focuses on optimizing a deep learning model for mobile phones, making pest counting more accessible and cost-effective for farmers. This is a different approach compared to the paper which uses an embedded device and cloud-based analysis. Both our work and the paper utilize deep learning techniques, and we may draw inspiration from their methodologies to inform and enhance our own approach.

2.1.3 Research Item 3

An evaluation of deep learning methods for small object detection by Nguyen et al. [5]

2.1.3.1 Summary of the research item

The paper dives into the difficulties of detecting small objects within the field of computer vision. A number of deep learning models were analyzed, namely YOLOv3[7],

RetinaNet [4], Fast RCNN[2], and Faster RCNN [8]. These models were tested on two datasets: a specific small object dataset, and subsets of PASCAL VOC dataset. Performance was measured based on accuracy, speed, and resource utilization. The paper found that two-stage models like Faster RCNN perform best in terms of accuracy, but are poorer in inference time. On the other hand, YOLOv3 is a good balance between accuracy and speed.

2.1.3.2 Critical analysis of the research item

The biggest strength of this paper lies in its investigation of small object detection by comparing various models. It presented thorough evaluations of models with differing architectures providing useful insights into their performance. Additionally, the paper also discusses trade-offs between accuracy and speed, which is another noteworthy strength. However, there are a few limitations and gaps. The paper mentions a drop in accuracy caused by intraclass similarity and variance, but does not discuss ways to overcome this. The paper also provides a comprehensive comparison of different models providing a range of insights such as inference time, RAM used in training/testing etc. but does not dive into the theoretical reasons why certain models excel in specific scenarios. Overall, the study is credible due to its clear methods in small object detection.

2.1.3.3 Relationship to the proposed research work

The findings of this paper are an important reference for our proposed research. Our focus is on detecting insects, which are essentially small objects. The comparative analysis of various deep learning models presented in the paper provides valuable insights into potential options for our work. It effectively compares different state-of-the-art models on their accuracies. Additionally, the insights on inference times and RAM usage are also relevant to our research, as they will help us optimize our model for mobile devices. We want to make sure our models are fast and resource efficient so they consume less energy on energy-constrained mobile phones. Moreover, the gap identified regarding intraclass similarity is also applicable to the different classes of insects we aim to detect. This highlights the need for us to address this issue in our own work.

2.1.4 Research Item 4

A survey of deep learning on mobile devices: Applications, optimizations, challenges, and research opportunities by Zhao et al. [11]

2.1.4.1 Summary of the research item

The paper reviews advancements in deploying deep learning models directly onto mobile devices rather than reliance on traditional cloud-based systems. It identifies key challenges faced in deploying complex neural networks onto resource-constrained mobile devices. In response to these challenges, it highlights optimization techniques and model conversion tools that can enhance model efficiency on mobile devices. Overall, the paper emphasizes the need for creating lightweight models and using resources efficiently, while also pointing out future research directions to tackle the current challenges in the field.

2.1.4.2 Critical analysis of the research item

The paper has a number of strong points. One is its detailed literature review, which helps readers understand the latest trends and challenges in mobile deep learning. It does a good job of categorizing different optimization approaches such as quantization and pruning. An example is how it discusses PruneFL, a pruning approach. The paper does a good job of covering many aspects, including accuracy-power trade-offs as well as inference times for models.

On the downside, the paper acts only as a survey for various methods. Due to this, it leans more on the theoretical side and does not include enough real-world applications.

2.1.4.3 Relationship to the proposed research work

The findings and methodologies presented in the paper are highly relevant to our proposed research. They emphasize the need to optimize deep learning models for mobile devices, which aligns with our current research objectives. The paper points out gaps in deployment of deep learning models on mobile devices due to resource constraints. Techniques such as quantization and pruning mentioned in the paper are essential for our own work.

Additionally, the paper discusses various accelerators and libraries available for mobile phones, as well as software and hardware optimization support. There are also insights into conversion tools for transferring models between frameworks. Furthermore, the paper's examination of inference time and power consumption provides valuable context for our project, especially regarding the accuracy-power trade-off, which is a key consideration in our approach. Overall, the insights from this paper will help shape our methodologies for model optimization as we dive into our research.

2.1.5 Research Item 5

Faster R-CNN and EfficientNet[9] for accurate insect identification in a relabeled yellow sticky traps dataset by Deserno et al. [1]

2.1.5.1 Summary of the research item

The paper discusses counting and classification of insects on yellow sticky traps using deep learning models. The authors in the paper work on a public bench-marking dataset which they first re-annotate due to mislabelling. The dataset contains three types of insect classes. Two main methods are applied: Faster RCNN with inception ResNet-v2[3], and EfficientNet. Data is augmented and split to improve model performance. Results show Faster R-CNN achieved weighted average accuracies of 92.51% and 81.27%, while EfficientNet-B3 reached an accuracy of 99.1% with high precision and recall rates. EfficientNet also took significantly less time to train compared to Faster RCNN.

2.1.5.2 Critical analysis of the research item

The paper has a number of strengths worth discussing. The biggest strength is how the authors provide extensive details regarding implementation, covering the entire process of data preparation, data preprocessing, model specifics, hyper-parameters, training/testing details etc. These details are followed by model comparisons on various metrics such as accuracy, precision, recall, training time, and number of epochs. The authors also re-annotate the dataset they use, keeping the newly labeled dataset public. Overall, the paper is highly practical.

However a number of gaps have also been identified. Due to the paper's highly practical approach, a lack in theoretical explanations behind the results is felt. Furthermore, the model comparisons in the paper do not include metrics such as inference times, nor do they address resource consumption in training/testing. Hence, trade-offs between speed and accuracy cannot be seen clearly. Lastly, the authors leave the problem of reducing false positives to future research works without delving into any strategies to counter this issue.

2.1.5.3 Relationship to the proposed research work

The paper is highly relevant to our proposed work since it focuses on the same task of counting and classifying different insect types on yellow sticky traps. The paper works with three different species of insect while our work focuses on species of canola flea beetle. The implementation details provided in the paper such as data pre-processing

techniques, hyperparameters, training/testing details will be particularly insightful for our own project. Additionally, the availability of a correctly labeled public dataset serves as a valuable benchmark for us. Some gaps do need to be addressed however, specifically regarding limitations of edge-devices such as mobile phones. The paper does not consider factors like energy consumption and inference times, which are crucial for optimizing our solution for mobile applications.

2.2 Analysis Summary of research items

Table 2.1 is a tabular representation of the research items with discussion of the critical analysis and relevance to proposed approach.

Research Item	Strengths	Limitations	Relevance to Proposed Work
Ullah et al. [10]	Custom FB-YOLO model, high mAP (90.16%), custom dataset for FBs, use of NMS-C algorithm	Expensive hardware, cloud reliance limits field deployment, integration challenges for mobile devices	Basis for our research, we aim to improve solution discussed in paper by bringing it to readily accessible mobile devices
Zhong et al. [12]	Good handling of low amount of data, YOLO for detection and SVM for species classification, uses yellow sticky traps for real-world data.	Limited dataset may affect model generalizability, still relies on cloud for processing.	Same goal of pest management as us, paper is one of the first in this domain using deep learning.
Nguyen et al. [5]	Comprehensive model comparison on YOLOv3, RetinaNet, Fast RCNN, and Faster RCNN. Evaluates accuracy, speed, resource use, insights on model architectures. Discusses power-accuracy trade-off	No solutions for intraclass similarity, lacks in-depth theoretical discussion.	Valuable insights for insect detection models, gives us a baseline of models to work with
Zhao et al. [11]	Focus on mobile devices, detailed optimization techniques (quantization, pruning), identifies resource constraints, software/hardware level optimizations	Theoretical focus, lacks practical applications and case studies, limited hands-on results	Related to our work since we aim to use mobile devices, details on optimizations insightful, relevant for addressing power efficiency and inference time

Research Item	Strengths	Limitations	Relevance to Proposed Work
DeSerno et al. [1]	Extensive implementation details such as data preparation, data preprocessing, model specifics, hyperparameters, training/testing details. Public dataset with re-annotation	Lacks theoretical explanations, no inference time/resource consumption metrics, does not address false positives	Highly relevant to our work due to similarity of work, insights into the practical details helpful for our own work. Public dataset acts as a benchmark for our work

Table 2.1: Literature Review Table

Chapter 3

Proposed Approach

This chapter outlines our proposed methodology to address the problems mentioned in the first chapter. Our primary goal is to deploy a deep learning system on mobile phones to count flea beetles on canola farms. Our approach experiments with several frameworks, algorithms, and techniques discussed in Chapter 2 to ensure the requirements of our research are met.

3.1 Research Methodology

We will work with the same dataset as identified in [10] while addressing its limitations related to expensive and relatively inaccessible specialized equipment. Our methodology includes experimenting with various data preprocessing techniques such as augmentation and image slicing to enhance model performance. We will choose our baseline models from our literature review, particularly from [12], [5] and [1] amongst others, and conduct experiments to refine model selection through parameter adjustments and architectural modifications. Additionally, we may need to explore alternative approaches, such as machine learning for classification as in the case of [12] and two-stage detection methods as in the case of [5]. Insights from the model implementation details in [1] will significantly aid us by providing us with practical details due to the similarity of our works. Furthermore, [1] will also provide us with a public benchmarking dataset for comparisons. We will compare the different models on various metrics such as accuracy (mean average precision), inference time, resource consumption, and energy consumption. After selecting the optimal model, our primary focus will shift to optimization where we will be utilizing techniques discussed in our reviewed literature, especially [11], which deals with mobile-specific optimizations. These optimizations aim to reduce model size and inference time while addressing accuracy-energy trade-offs essential for mobile devices which are constrained by battery. Extensive testing will be conducted, including eval-

uations on mobile platforms using frameworks like TensorFlow Lite and PyTorch Lite. Furthermore, we will employ Flutter for app development and utilize mobile conversion tools such as ONNX to ensure our solution is platform independent. By integrating these comprehensive methodologies, we aim to create an accurate and efficient deep learning model that can be easily integrated onto a mobile device rather than needing specialized equipment. Figure 3.1 outlines the workflow pertaining to our proposed solution.

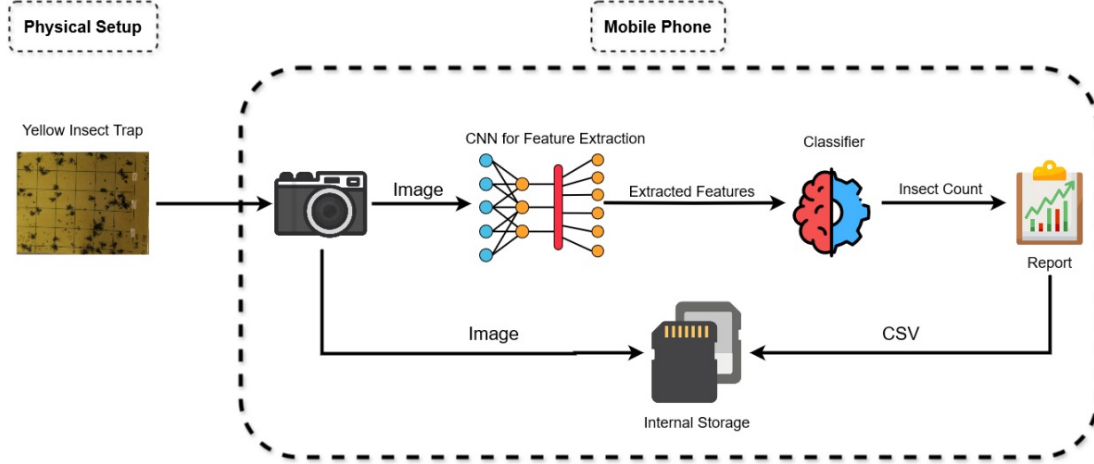


Figure 3.1: INS(P)ECT framework for automated FB detection and counting in canola crops using sticky traps

3.2 Research Steps

Our research consists of the following steps:

1. **Data Preprocessing:** Applying augmentation techniques such as rotation, flipping, scaling etc. as well as image slicing if necessary. Changes may need to be made to labeling/annotations.
2. **Model Training:** Training different models as identified in the literature review (e.g., YOLOv3, YOLOv8, Faster R-CNN, and EfficientNet) on the pre-processed data.
3. **Model Comparison:** Evaluating different deep learning architectures on various metrics such as accuracy (mean average precision), inference time, and energy consumption.
4. **Model Selection:** Choose the most effective model based on previous comparisons and experimentation.

5. **Model Optimization:** Making the models lightweight to better fit mobile devices using techniques discussed in our literature review.
6. **App Development and Model Integration:** Creating a cross-platform app and integrating our proposed optimized model.

Figure 3.2 outlines a detailed depiction of the steps involved in our research.

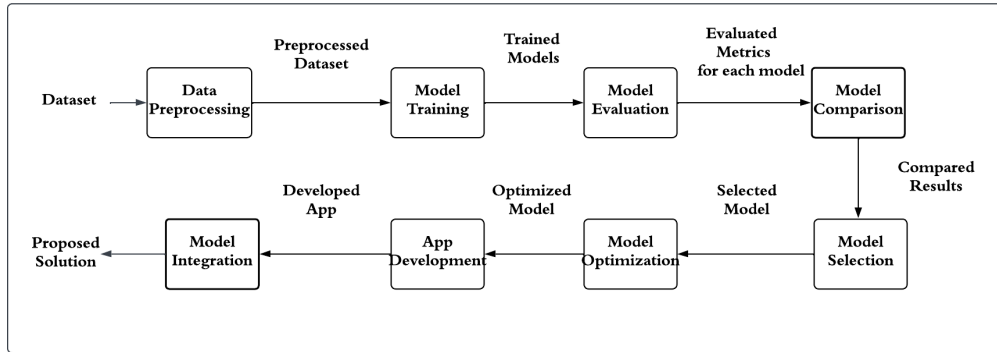


Figure 3.2: Flowchart of steps delineating the research

3.3 Conclusion

This chapter has outlined our proposed approach, combining a comprehensive methodology for developing a deep learning model optimized for mobile devices. It has also listed the steps we will take in our research to ensure our research objectives are met.

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